

THIEF RIVER FALLS, MN, USA

WMO: 727555

Lat: 48.067N

Lon: 96.183W

Elev: 1115

StdP: 14.11

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94956

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.1	-17.5	-27.7	1.2	-16.2	-26.0	1.3	-14.2	29.2	10.0	26.7	8.9	6.1	300	0.675

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	85.8	70.6	82.3	68.5	80.8	67.3	73.7	82.4	71.3	80.2	69.1	77.7	13.8	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.4	116.8	79.7	68.2	107.9	77.4	65.8	99.2	75.1	37.9	82.6	35.7	79.7	33.9	78.2	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
26.2	23.4	20.4		-27.4	91.2	6.3	2.9	-31.9	93.3	-35.6	95.1	-39.1	96.7	-43.7	98.8
			DB	-24.6	77.6	3.8	2.1	-27.3	79.1	-29.5	80.3	-31.7	81.4	-34.5	82.9
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.8	6.9	10.7	24.5	40.3	53.7	63.7	67.9	66.0	57.5	43.9	27.6	13.0
	DBStd	24.02	14.52	13.80	13.45	10.34	8.82	6.48	5.49	5.77	8.48	9.18	11.54	13.25
	HDD50	5692	1336	1100	792	317	63	1	0	0	25	235	675	1148
	HDD65	9514	1801	1520	1255	741	364	98	34	57	252	656	1123	1613
	CDD50	1967	0	0	2	26	176	412	556	496	251	46	2	0
	CDD65	313	0	0	0	0	12	59	125	88	27	2	0	0
	CDH74	2784	0	0	0	8	163	522	1026	772	270	23	0	0
	CDH80	644	0	0	0	1	38	119	241	180	61	4	0	0
	WSAvg	10.1	10.4	10.3	10.5	11.0	11.0	9.5	8.3	8.4	9.7	10.9	10.7	10.4
	PrecAvg	21.7	0.5	0.4	0.7	1.2	2.7	3.8	3.1	2.6	2.2	1.7	0.9	0.7
(15) Precipitation	PrecMax	28.3	1.1	1.3	2.2	2.7	6.5	8.2	5.6	6.6	5.6	4.3	3.2	1.5
	PrecMin	12.0	0.0	0.0	0.0	0.0	0.8	0.9	1.2	1.0	0.1	0.2	0.1	0.0
	PrecStd	3.8	0.4	0.3	0.6	0.6	1.4	1.7	1.2	1.3	1.3	1.1	0.8	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.3	42.9	62.8	73.0	84.2	88.5	89.8	90.1	86.4	76.9	58.9	39.4
		MCWB	34.2	39.2	53.5	55.4	64.9	71.7	73.2	73.6	70.3	60.8	49.9	35.4
	2%	DB	34.0	37.3	52.4	66.2	79.0	82.5	84.5	84.2	81.2	69.7	52.1	36.3
		MCWB	32.0	33.7	46.2	51.7	60.0	68.0	71.9	69.6	65.6	57.5	45.9	33.5
	5%	DB	30.0	34.5	45.7	62.6	73.2	79.9	82.1	81.4	76.8	63.2	46.4	33.8
		MCWB	28.7	32.0	40.6	49.1	57.6	65.6	70.1	67.9	63.5	53.6	41.6	31.6
	10%	DB	27.1	30.3	40.8	56.8	69.8	76.6	80.6	78.9	72.5	57.8	42.9	30.0
		MCWB	25.4	28.6	36.4	46.4	55.8	64.0	68.8	65.9	61.3	49.8	38.2	28.3
	0.4%	WB	35.0	39.1	54.2	58.2	67.7	74.7	77.1	75.2	72.0	62.8	51.7	37.0
		MCDB	37.4	42.3	58.7	69.3	78.8	83.4	85.4	85.0	82.6	72.5	55.3	39.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	31.9	34.5	46.2	52.9	63.7	71.0	74.5	72.2	68.4	58.8	46.2	33.6
		MCDB	33.5	37.2	51.5	63.9	73.5	80.2	82.5	82.0	77.6	67.9	50.2	35.8
	5%	WB	28.6	32.2	40.9	50.0	60.2	68.1	72.1	69.6	65.4	54.5	41.9	31.4
		MCDB	29.6	34.6	45.5	60.1	69.9	76.8	80.2	78.6	73.8	62.3	46.6	33.3
	10%	WB	25.5	28.7	37.0	47.1	57.4	65.8	69.9	67.7	62.8	50.5	38.4	28.4
		MCDB	27.4	30.4	40.5	56.3	66.6	74.0	77.9	75.8	70.7	57.4	42.1	29.6
	5% DB	MDBR	15.7	16.6	16.3	18.3	20.0	18.4	19.3	20.3	19.9	16.4	14.0	14.1
		MCDBR	15.4	15.2	19.7	26.7	26.4	22.9	22.1	23.5	25.7	24.3	19.3	15.0
		MCWBR	14.0	12.9	14.6	15.6	13.7	12.4	12.5	12.3	13.9	14.8	14.1	13.0
		MCWBR	14.9	14.2	19.4	24.5	22.4	19.4	20.1	21.2	21.8	22.1	18.4	15.0
(35) Mean Daily Temperature Range	5% WB	MCWBR	13.7	12.2	14.8	15.2	13.5	12.4	13.2	12.2	13.4	14.2	13.8	13.3
	taub		0.276	0.302	0.294	0.344	0.361	0.376	0.378	0.379	0.348	0.303	0.276	0.266
		taud	2.279	2.209	2.426	2.371	2.383	2.368	2.370	2.375	2.454	2.548	2.510	2.383
(42) Clear-Sky Solar Irradiance	Ebn at Noon		252	269	295	288	286	281	279	273	272	271	257	245
		Edh at Noon	23	32	30	36	36	37	37	35	30	23	19	19
	RadAvg		489	872	1236	1538	1729	1869	1991	1662	1221	747	438	341
(45) All-Sky Solar Radiation	RadStd		52	74	126	139	109	140	88	116	88	109	51	39

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (25 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page